Application Flow

Every journey in the product hangs off one object, the order, moving through six states while three audiences watch it: the guest, the kitchen and the floor. This tab gives the route map, each role's journey step by step, and what each screen must show in every state.

1. Route map

Route	Screen	Role	Notes
`/t/[qrToken]`	Guest menu	Guest	Entry point from QR
`/t/[qrToken]/cart`	Cart (sheet over menu)	Guest	
`/t/[qrToken]/orders`	Session orders and running bill	Guest	
`/o/[orderId]`	Order tracking	Guest	Canteen shows token
`/t/[qrToken]/pay`	Payment	Guest	
`/t/[qrToken]/receipt`	Receipt	Guest	
`/invalid`, `/closed`	Invalid QR, outlet closed or suspended	Guest	
`/login`, `/signup`, `/onboarding`	Staff auth and first-run setup	Staff	
`/app`	Home: today's KPIs and live summary	Owner, Manager	
`/app/orders`	Live orders board	Owner, Manager, Waiter	
`/app/tables`	Table tiles and QR management	Owner, Manager, Waiter	
`/app/menu`	Menu editor	Owner, Manager	

Route	Screen	Role	Notes
`/app/payments`	Payments and refunds	Owner, Manager	
`/app/analytics`	Reports	Owner, Manager	
`/app/staff`	Staff and PINs	Owner	
`/app/settings`	Outlet settings	Owner, Manager	
`/app/canteen`	Canteen live panel	Manager, Counter	Canteen outlets only
`/kds/[outletId]`	Kitchen screen	Kitchen	PIN or device login
`/display/[outletId]`	Pickup token display	Public	Canteen outlets only
`/admin`	Tenants and plans	Platform admin	

2. Order lifecycle across audiences

State	Guest sees	Kitchen sees	Dashboard sees	Next action by
`placed`	"Order sent. Waiting for confirmation." Cancel available	Nothing yet (or ticket, if auto-accept is on)	New order card with Accept and Reject; sound alert	Manager/Waiter, or auto
`accepted`	"Confirmed"	Ticket in New column with Start	Order under its table, Accepted	Kitchen
`preparing`	"Being prepared" with elapsed time	Ticket in Preparing with Mark ready	Preparing	Kitchen

State	Guest sees	Kitchen sees	Dashboard sees	Next action by
`ready`	Dine-in: "On its way". Canteen: "Ready. Collect with token 42" plus alert	Ticket in Ready, then clears	Ready, highlighted for the waiter or counter	Waiter/Counter
`served`	"Served. Enjoy." Pay prompt if unpaid	Gone	Served; table shows amount due	Guest pays or staff settle
`rejected` / `cancelled`	Reason shown; refund notice if paid	Ticket removed with a brief notice	Marked, with reason	None

3. Guest journey: dine-in

1. **Scan.** Camera opens `/t/[qrToken]`. Server validates the token, checks the outlet is open, opens or resumes the table session, sets the session cookie.
2. **Browse.** Menu loads with table label in the header. Guest scrolls categories, searches, filters by dietary tag.
3. **Configure.** Tapping an item with options opens the item sheet; required groups must be satisfied before Add is enabled. Simple items add in one tap.
4. **Review.** Cart bar shows count and total. Cart sheet shows lines, notes, bill summary.
5. **Place.** Tap Place order. Client sends lines with an idempotency key; server revalidates availability and recomputes totals; order is created as `placed`.
6. **Track.** Guest lands on `/o/[orderId]` and watches the status tracker update in realtime.
7. **Order more.** "Add more items" returns to the menu; new orders join the same session and bill.
8. **Pay.** From tracking or the orders page, guest taps Pay, chooses UPI or card, completes checkout, and returns to a confirmation. Or chooses pay at counter, which raises a bill request to staff.
9. **Receipt.** Receipt screen with download; optional phone number to receive it by message.
10. **Close.** Staff close the table; the session ends. Scanning again starts a fresh session.

Branches and edge cases

Situation	Behaviour
Invalid or regenerated QR	`/invalid` with "Ask staff for a new code"
Outlet closed or tenant suspended	`/closed` with opening hours
Item sells out before placing	Order rejected with the item named; cart highlights it for removal
Network drops while placing	Retry with the same idempotency key; never a duplicate
Order rejected by staff	Tracking shows the reason and a Back to menu action
Payment fails	Stay on the pay screen with Retry and Pay at counter
Payment succeeds but page closed	Webhook settles; next visit shows paid
Second guest scans the same table	Joins the same open session; sees all orders on the table
Table already has an open session from earlier guests	Staff close the stale session from the dashboard; guest sees a fresh one
Guest refreshes	Cart from local storage; session from cookie; tracking refetches

4. Guest journey: canteen

1. Scan the counter or poster QR (a canteen outlet's QR is not tied to a table).
2. Browse and build the cart as in dine-in.
3. Place order, then pay immediately; with the prepaid setting on, the order stays hidden from the kitchen until payment is captured.
4. Receive a token number on the tracking screen.
5. Watch status; at `ready` the screen alerts and the token appears on the pickup display.
6. Show the token at the counter; staff mark it handed over; the session closes automatically.

5. Kitchen journey

1. Open `/kds/[outletId]` on the kitchen device and sign in with a PIN; the device stays signed in.
2. Screen loads all active tickets, then subscribes to realtime.
3. A new accepted order appears with a sound and flash, oldest first.
4. Tap Start to move a ticket to Preparing; optionally tick items as they are done.
5. Tap Mark ready; the ticket moves to Ready and the waiter or guest is notified.
6. The ticket clears when marked served or handed over.

Situation	Behaviour
Connection lost	Banner "Reconnecting"; actions queue or disable; on reconnect, full refetch
Order cancelled mid-prep	Ticket shows Cancelled for 10 s, then clears
Ticket ageing	Border turns warning, then overdue, at outlet thresholds
Wrong tap	Undo for 5 s after each status change

6. Waiter and counter journey

1. Open `/app/orders` on a phone or tablet; sign in with a PIN.
2. New order alert: review, then Accept or Reject with a reason (skipped when auto-accept is on).
3. When an order turns Ready, deliver it and mark Served.
4. Respond to service requests (call waiter, bill request) and acknowledge them.
5. For cash, open the table, tap Settle, record cash received.
6. Close the table when the bill is fully paid; the table returns to free.

7. Owner and manager journey

First run

1. Sign up and verify.
2. Onboarding wizard: business details and GSTIN, first outlet (name, mode, hours, taxes), payment setup.
3. Build the menu: categories, then items with images, variants and add-ons.

4. Create tables; download and print the QR PDF.
5. Invite staff and set PINs.
6. Place a test order from a phone using the guided checklist on the home screen.

Daily

1. Home shows today's orders, revenue, average order value and anything stuck.
2. Live board during service; toggle item availability as stock runs out.
3. Handle refunds and cancellations from the order drawer.
4. Review analytics after service: top items, peak hours, prep times.

8. Canteen manager journey

1. `/app/canteen` shows the queue by token with paid status on the left and the selected order on the right.
2. Toggle availability from a quick panel during rushes.
3. Mark orders ready and handed over; the pickup display updates on its own.

9. Payment flow in detail

Step	Client	Server	Result
1	Tap Pay	Compute amount due from DB; create gateway order; insert payment `created`	Checkout parameters returned
2	Open gateway checkout; guest pays		
3	Post success signature	Verify signature; mark payment `authorized`	Fast UI confirmation
4		Webhook `payment.captured`: verify, mark `captured`, update session balance, emit `payment.updated`	Source of truth
5	Receive realtime event; show Paid		Receipt available
Failure	Show Retry	Webhook `payment.failed` marks `failed`	Balance unchanged

Step	Client	Server	Result
Refund	Staff tap Refund	Call gateway; on webhook mark `refunded`	Guest notified

10. Realtime sequence for one order

```
Guest            Server/DB              Kitchen            Dashboard
  | place order ---> |                      |                  |
  |                  | insert order(placed) |                  |
  |                  | -- order.created ----------------------> | alert
  | <- 201 order id  |                      |                  |
  | subscribe session channel               |                  |
  |                  | <------------- accept ------------------ |
  | <- order.updated | -- order.updated --> | new ticket       |
  |                  | <-- start ---------- |                  |
  | <- preparing     | -- order.updated ----------------------> |
  |                  | <-- ready ---------- |                  |
  | <- ready         | -- order.updated ----------------------> | highlight
  |                  | <------------- served ------------------ |
  | <- served        |                      | ticket clears    |
```

11. Screen states

Every screen implements these states; none may be left blank.

Screen	Loading	Empty	Error	Offline
Guest menu	Skeleton cards	"Menu coming soon"	Retry	Cached menu if available; ordering disabled
Cart	n/a	"Your cart is empty" with Browse menu	Inline line errors	Place order disabled with reason
Tracking	Skeleton stepper	n/a	Retry	Reconnecting banner; last known status
KDS	Skeleton tickets	"No active orders"	Full-screen retry	Reconnecting banner
Orders board	Skeleton groups	"No orders yet"	Retry	Reconnecting banner

Screen	Loading	Empty	Error	Offline
Menu editor	Skeleton rows	"Add your first category"	Toast, form preserved	Saves disabled
Analytics	Skeleton charts	"No data for this range"	Retry	n/a

12. Notifications

Event	Recipient	Channel
New order	Dashboard, then kitchen once accepted	In-app sound and visual
Order accepted, preparing, ready, served	Guest	In-app realtime; optional message at ready in canteen mode
Order rejected or cancelled	Guest, kitchen	In-app
Service request	Dashboard	In-app sound and badge
Payment captured or failed	Guest, dashboard	In-app
Order waiting too long	Dashboard	In-app warning

13. Permissions at the screen level

Screen	Owner	Manager	Waiter/Counter	Kitchen
Home, Analytics	Yes	Yes	No	No
Orders board	Yes	Yes	Yes	No
Tables	Yes	Yes	View and settle	No
Menu editor	Yes	Yes	Availability only	No
Payments and refunds	Yes	Yes	Record cash only	No
Staff	Yes	View	No	No
Settings	Yes	Own outlet	No	No

Screen	Owner	Manager	Waiter/Counter	Kitchen
Kitchen screen	Yes	Yes	No	Yes

Unauthorised routes redirect to the role's home screen; the server enforces the same rules regardless of what the UI hides.